

Practice 5: OVB's Hide and Seek

25117 - Econometrics

Universitat Pompeu Fabra

November 28th, 2023

What we learned in the last lesson

- When Y is a binary variable, the population regression function shows the probability that $Y = 1$ given the value of the regressors, $X_1, X_2, \dots, X_k$.
- The linear multiple regression model is called the linear probability model when Y is a binary variable because the probability that $Y = 1$ is a linear function of the regressors.
- Probit and logit regression models are nonlinear regression models used when Y is a binary variable. Unlike the linear probability model, probit and logit regressions ensure that the predicted probability that $Y = 1$ is between 0 and 1 for all values of X .
- Probit regression uses the standard normal cumulative distribution function. Logit regression uses the logistic cumulative distribution function. Logit and probit coefficients are estimated by maximum likelihood.
- The values of coefficients in probit and logit regressions are not easy to interpret. Changes in the probability that $Y = 1$ associated with changes in one or more of the X 's depends on the initial value of X .
- Hypothesis tests on coefficients in the linear probability, logit, and probit models are performed using the usual t - and F -statistics.

Practice Setting

Your Research Question

What is the elasticity-price of cigarettes?

Why are we interested in knowing the elasticity of demand for cigarettes?

- **Non-monetary externalities** — role for government intervention to discourage smoking health effects of second-hand smoke?
- **Monetary externalities** — Pressure on public health system

By the way... what is an externality?

Threats to Internal Validity

Question

What is internal validity? How does it differ from external validity?

Remember... There are 5 threats to the internal validity of regression studies:

- Omitted variable bias
- Wrong functional form
- Errors-in-variables bias
- Sample selection bias
- Simultaneous causality bias

All of these imply that $\mathbb{E}(u_i | X_1, X_2, \dots, X_k) \neq 0$ (or that the conditional mean independence fails) – in which case OLS is biased and inconsistent!

Question

What can we do to tackle each one of them?

Panel Dataset

Question

What is a panel dataset? Why is it useful? A cross-sectional dataset?

Your panel:

- 48 continental U S states, 1985–1995
- Annual cigarette consumption
- Average prices paid by end consumer (including tax)
- Personal income
- Tax rates (cigarette-specific and general statewide sales tax rates)

Question

What would be your main empirical strategy to answer the research question at hand?

Estimation Strategy

We need to combine both IV and panel data methods!

- We need unit fixed effects which control for unobserved state-level characteristics that affect the demand for cigarettes and the tax rate, as long as those characteristics don't vary over time.

$$\log(Q_{it}^{cigarettes}) = \alpha_i + \beta_1 \log(P_{it}^{cigarettes}) + \beta_2 \log(Income_{it}) + u_{it}$$

Question

How should you interpret β_1 ? Why controlling for *Income*?

- We need IV methods to handle the simultaneous causality bias that arises from the interaction of supply and demand.

Question

What is simultaneous causality bias and how is it related to the problem at hand?

Long Term Effects

One way to account for state fixed effects is to model long-term effects (why?), e.g., consider 10-year changes, between 1985 and 1995

Rewrite the regression in “changes” form:

$$\begin{aligned} \log(Q_{i,1995}^{cigarettes}) - \log(Q_{i,1985}^{cigarettes}) &= \beta_1 [\log(P_{i,1995}^{cigarettes}) - \log(P_{i,1985}^{cigarettes})] \\ &+ \beta_2 [\log(Income_{i,1995}) - \log(Income_{i,1985})] \\ &+ [u_{i,1995} - u_{i,1985}] \end{aligned}$$

Then estimate the demand elasticity by 2SLS using 10-year changes in the instrumental variables

Question

How would you write the corresponding short-run specification, while accounting for state fixed effects, α_i ?

Long Term Effects

	$\log(Q_{i,1995}^{cigarettes}) - \log(Q_{i,1985}^{cigarettes})$		
	(1)	(2)	(3)
$\log(P_{i,1995}^{cigarettes}) - \log(P_{i,1985}^{cigarettes})$	-0.094 (0.21)	-1.34 (0.23)	-1.20 (0.20)
$\log(Income_{i,1995}) - \log(Income_{i,1985})$	0.53 (0.34)	0.43 (0.30)	0.46 (0.31)
Constant	-0.12 (0.07)	-0.02 (0.07)	-0.05 (0.06)
Instrumental Variable(s)	Sales tax	Cigarette tax	Both
First-stage-F-statistic	33.7	107.2	88.6
J-Statistic	-	-	4.93 [0.026]

Robust standard errors are reported in parenthesis. p-values are reported in brackets.

Pop-up Quiz

Question(s)

1. Interpret the coefficients of interest
2. Do they seem large? Small?
3. How do we interpret the coefficients of the LR difference in income? What does it suggest?
4. What are the two conditions required for a valid instrument?
5. How to check for each condition?
6. What is a J-test?
7. How do we interpret the J-statistic?
8. Here, we are looking at long-run (ten-year change) elasticities. What would you expect a short-run (one-year change) elasticity to be — more or less elastic?
9. What are the remaining threats to internal validity?
10. Discuss the external validity of the results

Good Luck for the Exam!